Kendall Sullivan

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Objective

As a student, researcher, and educator, I aim to equip all learners with high utility tools they can harness to take agency over their learning, improve their learning efficiency, and effectively retain and utilize their knowledge in innovative ways. My current objective is to evaluate the efficacy and effectiveness of empirically-grounded educational technologies for the purpose of improving learners' ability to transfer and utilize their knowledge while retaining agency over the learning process. In doing so, I aim to contribute to the collaborative effort of reducing the equity gap in education by expanding access to effective tools that adapt to individual learner needs.

Education

B.A., Psychology with Honors, University of California, Riverside, *Sept 2023 - Expected June 2027*
- Cumulative Graduate GPA: 4.00
- Cumulative Undergraduate GPA: 3.98

High School Diploma, University High School, Irvine, CA, *Aug 2019 - June 2023*
- Cumulative GPA: 3.98

Independent Research Projects

Evaluating the utility of GenAI as a retrieval practice tool: A comparison of student-, AI-, and experimenter-generated testing
Honors Capstone Research Project, University of California, Riverside | May 2025 - Present
Faculty Advisor: Dr. Annie Ditta, Ph.D.

- Evaluated the effects of testing type (student-generated, AI-generated, experimenter-generated) and question type (previously tested, previously not tested) on final test performance and students' learning perceptions.
- Designed, developed, and conducted a series of remote experiments using mock-classroom learning instruction.
- Trained and managed a team of research assistants; taught them how to proctor pre-recorded lectures, learning assessments, and questionnaires via Zoom with screen share.
- Developed a data cleaning and analysis workflow in R using machine learning to automate the coding process and minimize human error and researcher bias in coding.
- Drafted the introduction and method for the manuscript; intend to complete and submit the manuscript to an educational psychology journal; registered to present the findings at the Annual Meeting of the Psychonomic Society.

Investigating the relationship between cognitive load, comprehension, and flexible transfer in multimedia statistics learning: A classroom study
Independent Research Project, University of California, Riverside | May 2026 - Present
Faculty Advisor: Dr. Annie Ditta, Ph.D.

- Designed and developed a classroom study to translate the two-experiment research project, Sullivan & Ditta (ready to submit), to an ecologically valid instructional context.
- Assessed 1) whether a quasi-manipulation of perceived extraneous cognitive load significantly predicts comprehension and transfer performance on a college statistics exam and 2) whether

individual differences in perceived intrinsic, extraneous, germane, and total cognitive load significantly predict those same learning outcomes.
- Wrote an IRB protocol and two preregistrations.

Investigating the relationship between cognitive load, comprehension, and flexible transfer in multimedia statistics learning
Independent Research Project, University of California, Riverside | Sept 2024 - June 2026
Faculty Advisor: Dr. Annie Ditta, Ph.D.

- Systematically evaluated the effect of low vs. high cognitive load on comprehension and flexible, content-grounded transfer of knowledge in an applied STEM learning context; assessed whether comprehension mediated any cognitive load effect on transfer.
- Designed, developed, and conducted two laboratory experiments using mock-classroom instruction.
- Trained and managed a team of research assistants to help with data collection.
- Developed a data cleaning and analysis workflow in R for a mediation path model fitted with negative binomial regression to account for the over-dispersed, zero-inflated count outcome.
- Used machine learning to efficiently code qualitative data and minimize human error and researcher bias in coding.
- Published all study materials and R code on a ResearchBox repository; will be made public once the manuscript is accepted.
- Drafted a manuscript for submission to a psychology research journal; learned how to effectively communicate my work to a lay audience through multiple oral and poster presentations.
- Drafted a second manuscript for a secondary data analysis re-evaluating the factor structure of the cognitive load measure utilized in the experiments through a model comparison of two confirmatory factor analysis models.

Exploring the relationship between comprehension and functional creativity in multimedia statistics learning
Independent Research Project, University of California, Riverside | Sept 2024 - Dec 2024
Faculty Advisor: Dr. Annie Ditta, Ph.D.

- Conducted a quasi-experiment assessing the relationship between comprehension and functional creativity in multimedia statistics instruction.
- Developed and implemented a coding scheme to quantity functional creativity.
- Drafted a manuscript; presented a poster.

Research Grants

UCR Research Mini Grant, 2026 - $1,000

UCR Travel Mini Grant, 2026 - $1,500

UCR MSRIP Research Grant, 2025 - $500

Research Publications

Sullivan, K. E., Trego, M. M., & Ditta, A. S. (in preparation). Evaluating the utility of GenAI as a retrieval practice tool: A comparison of student-, AI-, and experimenter-generated testing.

Sullivan, K. E., & O'Rourke, H. P. (submitted). Re-evaluating the factor structure of cognitive load subtypes in undergraduate students. *Psychological Assessment.*

Sullivan, K. E., & Ditta, A. S. (ready to submit). Investigating the relationship between cognitive load, comprehension, and flexible transfer in multimedia statistics learning. *Psi Chi Journal of Psychological Research.*

Research Conference Papers

Sullivan, K. E., & Ditta, A. S. (in preparation). How individual differences in cognitive load predict comprehension and transfer performance on a college statistics exam.

Sullivan, K. E., & Ditta, A. S. (in preparation). Investigating the relationship between cognitive load, comprehension, and transfer in multimedia statistics learning: A classroom study.

He, J. J., & **Sullivan, K. E.** (submitted). From prior conceptions to new understanding: tracing how an AI literacy game shapes middle school students' understanding of AI. *Proceedings of the International Conference of the Learning Sciences.*

Research Manuscripts Not Submitted

Sullivan, K. E., & Ditta, A. S. (2026). Fine-tuning the investigation of the relationship between cognitive load, comprehension, and flexible transfer in multimedia statistics learning. *Not Submitted.*

Sullivan, K. E., Cruz-Hernandez, F. A, Onsing, J., & Ditta, A. S. (2024). Exploring the relationship between comprehension and functional creativity in multimedia statistics learning. *Not Submitted.*

Research Conference and Symposia Presentations

Sullivan, K. E., Trego, M. M., Ditta, A. S. (in preparation). Evaluating the utility of GenAI as a retrieval practice tool: A comparison of student-, AI-, and experimenter-generated testing using scaffolded question generation, Experiment 2. Committed by grant contract to present a poster at UCR's Annual Undergraduate Research and Creative Activities Symposium (UGRS) in Spring 2027.

Sullivan, K. E., Trego, M. M., Ditta, A. S. (in preparation). Evaluating the utility of GenAI as a retrieval practice tool: A comparison of student-, AI-, and experimenter-generated testing, Experiment 1. Accepted to present a poster at the Annual Meeting of the Psychonomic Society in Fall 2026.

Sullivan, K. E., Ditta, A. S. (2026). Investigating the relationship between cognitive load, comprehension, and transfer in multimedia statistics learning. Presented a poster at UCR's Annual Undergraduate Research and Creative Activities Symposium (UGRS) and at UCR's Annual Undergraduate Psychology Research Conference (R'PSYC) in Spring 2026. https://drive.google.com/file/d/1N8uzc4k7lITgn2_g3RHjl9t2YxxmLxnD/view?usp=sharing

Sullivan, K. E., Ditta, A. S. (2025). Reducing cognitive load can help you remember, but can it make you more creative? An investigation into the effect of cognitive load on comprehension and creativity in statistics learning. Presented a poster at UCR's Annual Undergraduate Psychology Research Conference (R'PSYC) in Spring 2025. https://drive.google.com/file/d/1WHgJ8Ymx6DsAz_rgPuKrkDWFdyhxkgLz/view?usp=sharing

Sullivan, K. E., Cruz-Hernandez, F. A., Ongsing, J., Ditta, A. S. (2025). Exploring the relationship between comprehension and creativity in statistics learning. Presented a poster at UCR's Annual Undergraduate Research and Creative Activities Symposium (UGRS) in Spring 2025. https://drive.google.com/file/d/1yqIyYEzhXWpLpvE7ofR6vfQen4axJDM7/view?usp=sharing

Non-Refereed Research Symposia Talks

Sullivan, K. E. (2026). The Benefits and Pitfalls of Data Cleaning with Machine Learning. Presented a talk at UCR Psychology's Quantitative Methods Proseminar in Spring 2026.

Sullivan, K. E. (2026). Optimizing Coding Efficiency. Presented a talk at UCR Psychology's Quantitative Methods Proseminar in Winter 2026.

Sullivan, K. E. (2025). Challenging Statistical Status Quo. Presented a talk at UCR Psychology's Quantitative Methods Proseminar in Fall 2025.

Sullivan, K. E., Trego, M. M., Ditta, A. S. (2025). Evaluating the utility of GenAI as a practice testing tool: A comparison of student-, AI-, and experimenter-generated testing, Research Proposal Presentation. Presented an emerging research talk at UCR's Mentoring Summer Research Internship Program (MSRIP) Symposium in Summer 2025.

Paid Research Positions

Collaborative Researcher, AI Intervention Project, PI: Dr. Mariam Salloum, June 2026 - Sep 2026
Collaborated with computer scientists on an AI educational intervention classroom study; contributed education and psychology perspectives. Refined the study design and defined the research questions. Conducted literature review and synthesis; extracted and adapted validated survey items for quantitative and qualitative assessment in the current project assessing differences in computer science students' perceptions and use of GenAI for coding pre- and post-intervention. Developed data cleaning and analysis plans and workflows in Google Colab; conducted data analysis; drafted project reports.

Independent Researcher, Mentoring Summer Research Internship Program (MSRIP), PI: Dr. Annie Ditta, June 2025 - Aug 2025
Designed, developed, and launched a remote experiment using mock-classroom learning instruction; received a $4500 stipend to pursue this research through a full-time, 8-week summer research internship. Actively participated in graduate school preparation workshops; learned how to craft research statements. Evaluated the utility of GenAI as a retrieval practice tool through a comparison of the effects of student-, AI-, and experimenter-generated testing on final test performance and students' learning perceptions. Expanded this multi-experiment research project into my University Honors Capstone Project.

Research Assistantships

Research Assistant, Stanford University DISTAL Lab with Dr. Victor R. Lee, July 2026 - Present
Collaborated with a doctoral student to iteratively develop a coding scheme and analysis plan for interview transcript data using thematic analysis. Scrutinized and identified patterns in behavioral data through several rounds of coding. Developed and executed a plan for quantifying and analyzing qualitative data. Co-authored the resulting write-up.

Research Assistant, UCR TeCD Lab with Dr. Mariam Salloum, Apr 2026 - Sept 2026
Refined study design and defined research questions. Conducted literature review and synthesis; extracted and adapted validated survey items for quantitative and qualitative assessment in the current project. Developed coding workflows in Google Colab for pilot data analysis.

Research Assistant, UCR TEACH Lab with Dr. Annie Ditta, July 2024 - Aug 2025
Ran participants through mock-classroom learning instruction and testing. Proctored pre-recorded lectures, learning assessments, and questionnaires. Actively participated in journal club discussions and presented my work at weekly lab meetings.

Research Assistant, UCR Life Events Lab with Dr. Kate Sweeny, July 2024 - Mar 2025
Ran participants through observational studies with EKG recording; administered personality assessments. Coded behavioral data in multimedia videos of adolescents and university students. Assessed participants on impatience, boredom, anger, and excitement.

Research Assistant, UCR Social Neuroscience Lab with Dr. Brent Hughes, Apr 2024 - Aug 2024
Ran participants through survey-based studies on memory mapping; administered personality assessments. Assessed participant eligibility for fMRI studies.

Academic Awards and Honors

Dean's Honor List, University of California, Riverside
Made the Dean's Honor List every eligible quarter from Fall 2023–Present (i.e., each quarter I was enrolled in at least 12 undergraduate or 8 graduate GPA-units with no grade below a "B" at an average GPA of 3.5 or higher).

Stella R. Arambel Academic Excellence Award
Awarded this faculty-nominated honor for outstanding achievement and promise in a career in psychological research at UCR's Annual Senior Celebration in Spring 2026.

Maria Franco-Gallardo Excellence Award
Awarded this honor, along with a $250 prize, for 'best oral presentation' at UCR's Mentoring Summer Research Internship Program (MSRIP) Symposium, 2025.

Psi Chi Exemplar Member Award
Nominated and voted for by my peers in UCR's chapter of the Psi Chi Honor Society in Psychology in Spring 2025 for performing as an exemplar of what Psi Chi students strive to embody.

Teaching Experience

Course Designer, Developer, and Facilitator, UCR R'Course, Apr 2026 - Present
Cognitive Principles of Teaching and Learning Undergraduate Seminar
Faculty Mentor: Dr. Hayden Hendley, Ph.D.

Designed and developed a syllabus for an undergraduate seminar on the cognitive principles of teaching and learning to facilitate as the lead instructor. Selected as one of only seven UC Riverside undergraduates in the Spring 2026 R'Course application cycle to develop and facilitate the course as the lead instructor during the 2026–2027 academic year.

Peer Tutor, Self-Organized, Jan 2024 - Mar 2024
Tutored peers at UC Riverside in PSYC 011 - Psychological Methods: Statistical Procedures. Tailored tutoring approaches to each student's pain points in applied statistics, then guided them to master their specific areas of difficulty. Helped students develop analytical and transfer skills.

Private Tutor, Self-Employed, Aug 2021 - Sept 2023
Tutored students in every level of math from fourth to ninth grade. Tailored tutoring approaches to each student's unique personality and their respective pain points in math, then guided them to master their specific areas of difficulty. Helped students develop problem-solving skills.

Peer Tutor, University High School Academic Student Coalition, Sept 2019 - May 2023
Tutored peers at University High School from freshman to senior level math, ranging from Algebra 1 to AP Statistics, in addition to AP Macroeconomics, CP Physics, AP Physics 1, and AP Physics 2. Facilitated private tutoring sessions four to five times a week; worked with students for hour-long increments on a need basis. Broke down concepts into their fundamental units to make the material digestible for struggling students.

Other Academic Activities

Official Member (Feb 2025 - Present), Prospective Member (Oct 2023 - Feb 2025), Psi Chi International Honor Society in Psychology
Regularly attended and actively participated in general meetings and discussions on topics in psychology. Developed critical and creative thinking skills for developing research projects with mentorship from an upperclassman member in the selective mentorship program.

Quantitative Methods Proseminar Attendee, UCR Psychology, Apr 2025 - Present
Attended weekly in-depth quantitative methods proseminars through the Department of Psychology at the University of California, Riverside with faculty, postdoctoral researchers, and doctoral researchers. Presented my work each quarter of my third year of undergrad and solicited feedback on streamlining my data coding and analysis plans for my presented projects. Actively engaged in scholarly methods discussions of other doctoral students' work.

Cognition and Cognitive Neuroscience Proseminar Attendee, UCR Psychology, Sept 2024 - Present
Attended weekly in-depth cognition and cognitive neuroscience proseminars through the Department of Psychology at the University of California, Riverside with faculty, postdoctoral researchers, and doctoral researchers. Actively attended to scholarly discussions of doctoral students' and visiting faculty members' work.

Memory Meetings Attendee, UCR Mnemonists, Aug 2025 - July 2026
Attended monthly memory meetings with the mnemonists group at the University of California, Riverside composed of faculty, postdoctoral researchers, graduate student researchers, and myself in the cognition and cognitive neuroscience area of the Department of Psychology. Discussed research on learning and memory through a neuroscience lens; actively attended to faculty discussion in workshopping sessions for developing RO1 grant proposals.

Board Member, Psi Chi International Honor Society in Psychology, June 2024 - June 2025
Organized and facilitated general meetings, research workshops, and discussions on topics in psychology as the community promoter on the board of UCR's chapter of Psi Chi. Developed and facilitated research workshops during which I shared resources for undergraduates interested in getting involved in research. Moderated live panel question and answer sessions during research workshops with experienced researchers.

University Honors Program Study Group Leader, Self-Organized, Oct 2023 - June 2024
Organized and facilitated discussions and review sessions for fellow UCR honors students in my cohort. Generated study outlines and led discussions on course content; encouraged student participation.

Other Non-Academic Activities

Cardio and Yoga Practitioner, Sept 2023 - Present
Woke/wake up at 4:55am everyday without fail and began/begin every morning with self-directed cardiorespiratory fitness training and yoga. Never took a rest day, despite injury, illness, or insomnia. Flexibility was exercised in terms of duration of each activity on the basis of other life demands, such as academic course schedule, research conferences, and final exams with a priori preparation.

Cardio, Cycling, and Yoga Practitioner, June 2023 - Sept 2023
Ran, cycled and practiced yoga daily. Took a single rest day due to illness. Continued to push through the pain on a daily run, cycle, and yoga practice despite two notable injuries.

Graduate Courses Enrolled

EDUC 230M: Multimedia and AI in STEM Education (Fall 2025)

PSYC 207C: Processes of Cognitive Development (Spring 2026)

PSYC 211: Quantitative Methods in Psychology (Fall 2025)

PSYC 212: Multiple Regression and ANOVA (Winter 2026)

PSYC 214: Structural Equation Modeling (Spring 2026)

PSYC 270: Current Research in Quantitative Psychology (Fall 2025, Winter 2026, Spring 2026)

PSYC 271: Cognitive Principles of Teaching and Learning (Winter 2026)

Graduate Courses Audited

PSYC 203B: Cognition and Cognitive Neuroscience Core (Winter 2026)

PSYC 259: Mediation Analysis (Winter 2026)

PSYC 270: Current Research in Quantitative Psychology (Spring 2025, Fall 2026)

PSYC 283: Proseminar on Current Research in Cognitive Psychology (Fall 2024, Winter 2025, Spring 2025, Fall 2025, Winter 2026, Spring 2026, Fall 2026)

PSYC 285: Proseminar on Current Research in Social/Personality Psychology (Fall 2026)